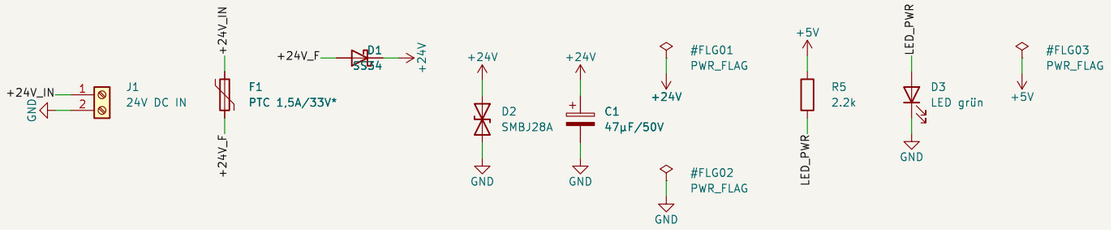
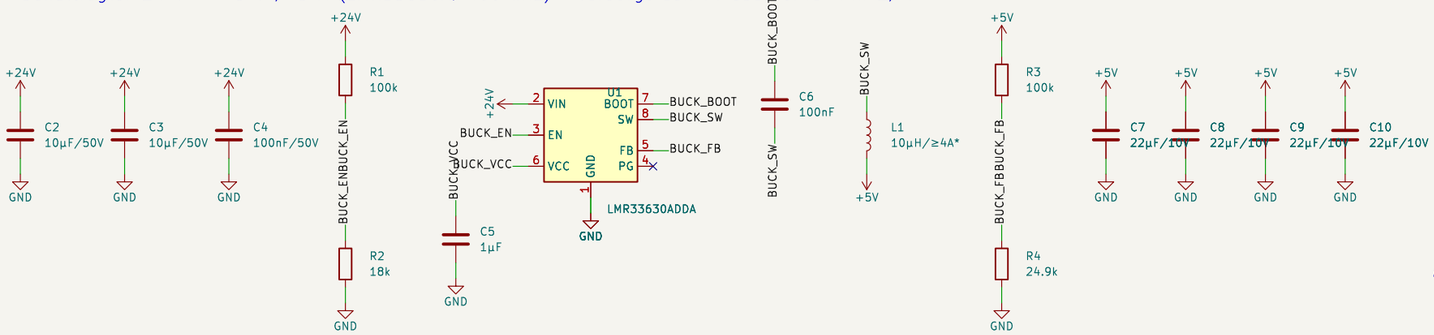
HETA Smart Filter Monitoring - Raspberry-Pi-HAT (Entwurf v0.1)

Ersetzt AnoPi-Shield, externen DC/DC-Wandler und fliegende Verdrahtung. Werte mit * vor Fertigung gegen Datenblatt prüfen (siehe hardware/heta-hat/README.md).

1 24-V-Eingang: Sicherung, Verpolschutz, Überspannungsschutz

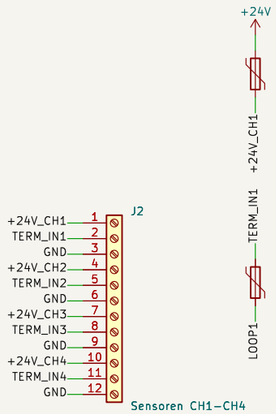


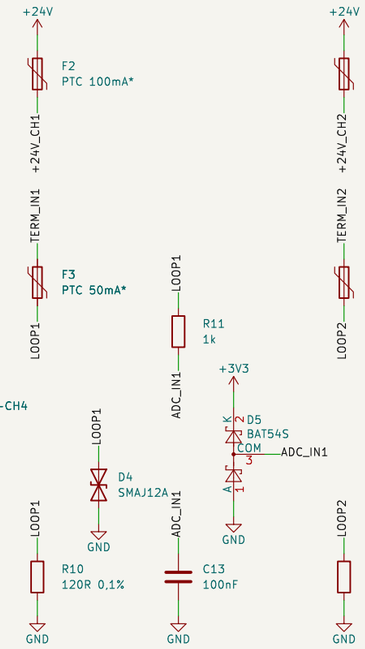
2 Schaltregler 24 V -> 5 V / 3 A (LMR33630A, 400 kHz) - versorgt den Pi über GPIO-Pin 2/4



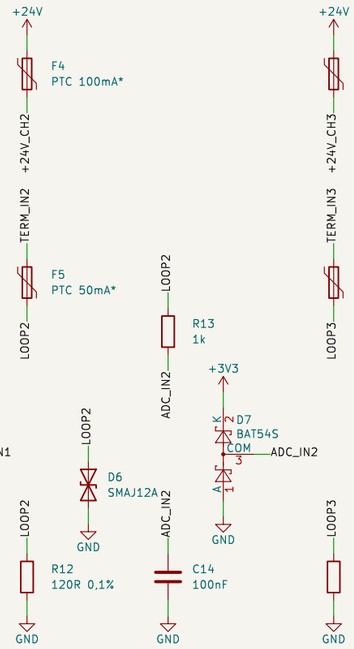
7 4 x 4-20-mA-Eingang (2-Leiter: +24V/IN, 3/4-Leiter: IN/GND) - Bürde 120 Ohm: 20 mA = 2,40 V

Kanal 1: p1 Eintrittsdruck

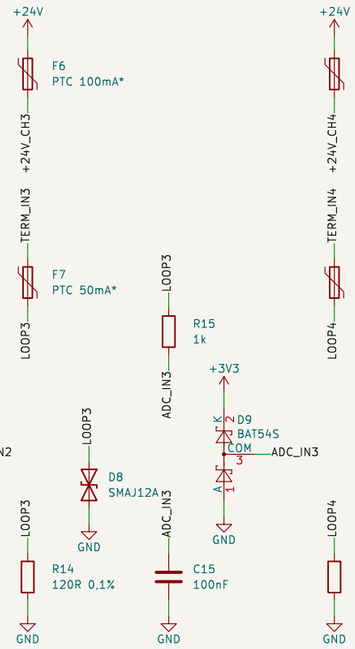




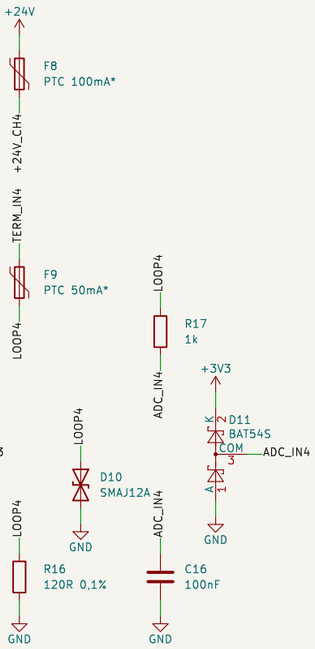
Kanal 2: p2 Austrittsdruck



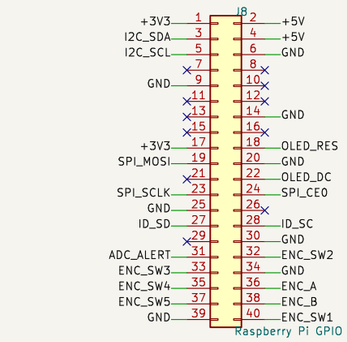
Kanal 3: T Temperatur



Kanal 4: Q Durchfluss

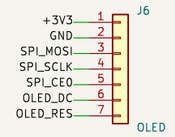


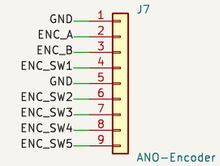
3 Raspberry-Pi-GPIO (40-pol., HAT)



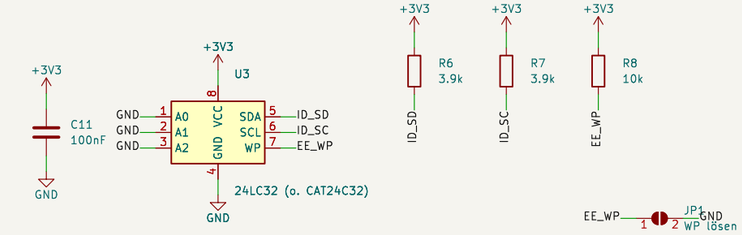


5 OLED und Encoder

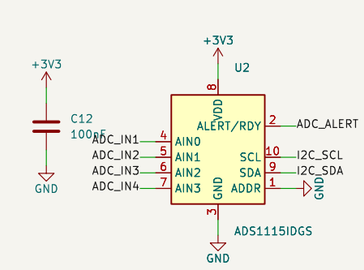




4 HAT-ID-EEPROM (Raspberry-Pi-HAT-Spezifikation)



6 ADC 16 Bit, I2C 0x48 (ADS1115)





Nicht freigegeben - Werte mit * gegen Datenblatt prüfen
4 x 4-20 mA (ADS1115), 24 V -> 5 V / 3 A, HAT-EEPROM, OLED + Encoder

HETA

Sheet: /
File: heta-hat.kicad_sch

Title: HETA Smart Filter Monitoring - Raspberry-Pi-HAT

Size: A3	Date: 2026-10-29	**Rev: 0.1 (Entwurf)**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/1